CHAPTER 9

High Impact Database Maintenance

You've seen how row modifications are made up of tuples, how there can be live and dead tuples, and how this is part of the Multiversion Concurrency Control (MVCC[1]) system.

With this knowledge, you're ready to learn about database maintenance in greater depth and see how automatic maintenance is performed and where you can help out by manually conducting maintenance.

For databases with hundreds of millions of rows and a high volume of queries you'll need to tune your system and possibly conduct manual maintenance operations.

You'll learn about the "VAR" aspects of database maintenance, which are Vacuum, Analyze, and Reindex.

Maintenance Terminology



- Visibility Map — Tracks which row versions (tuples) are visible to transactions
- Vacuum — Mark space as reusable, reclaim space (Vacuum Full), update Visibility Map
- Analyze — Update statistics about table data
- Reindex — Rebuild an index

First you'll get an introduction or refresher on Autovacuum.

1. https://www.postgresql.org/docs/current/mvcc.html